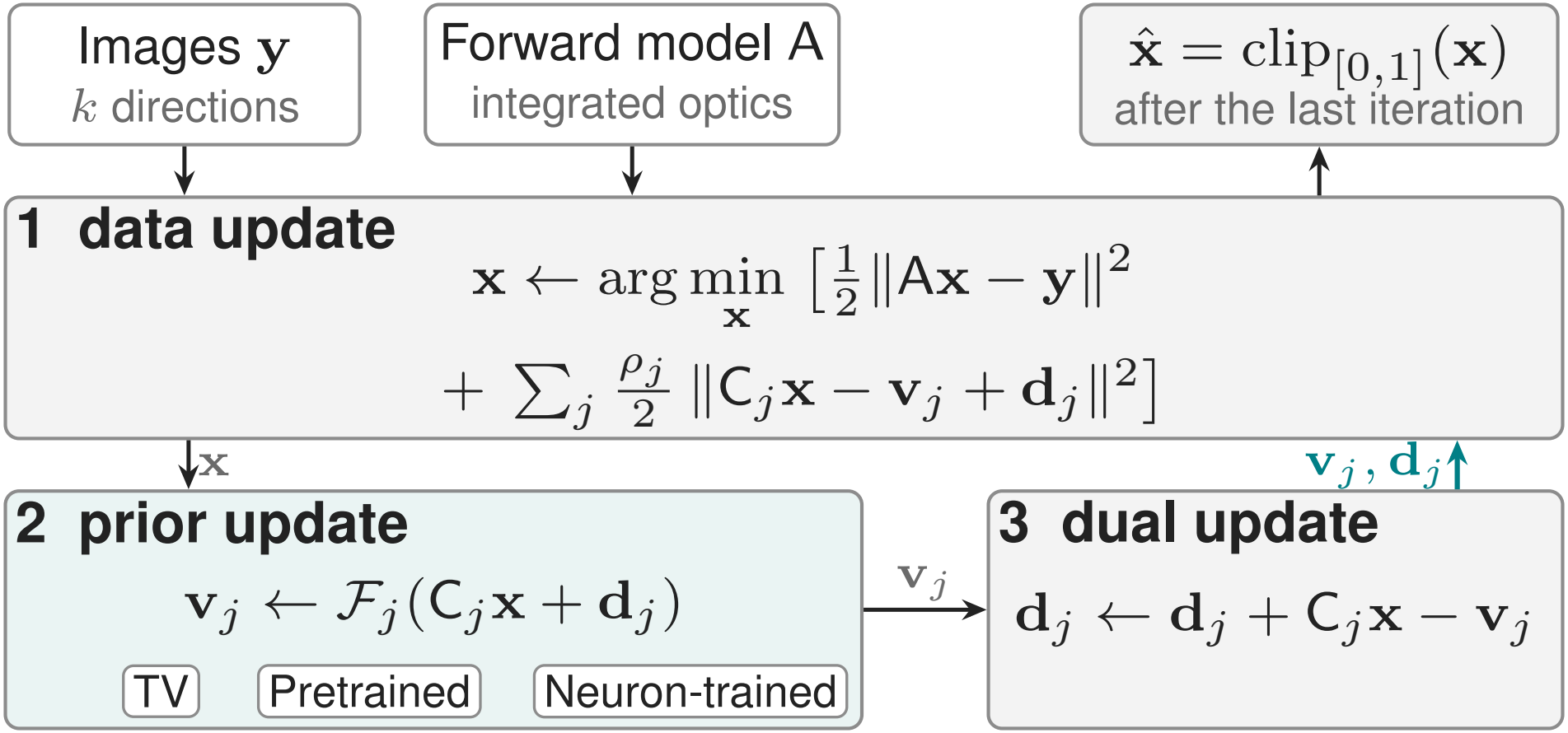

Images $\mathbf{y}$
k directions
Forward model A
integrated optics
$\hat{\mathbf{x}} = \mathrm{clip}_{[0,1]}(\mathbf{x})$
after the last iteration
1 data update
$\mathbf{x} \leftarrow \arg\min_{\mathbf{x}} \left[\frac{1}{2} \|\mathrm{A}\mathbf{x} - \mathbf{y}\|^2 + \sum_j \frac{\rho_j}{2} \|\mathrm{C}_j\mathbf{x} - \mathbf{v}_j + \mathbf{d}_j\|^2 \right]$
$\mathbf{x}$
$\mathbf{v}_j, \mathbf{d}_j$
2 prior update
$\mathbf{v}_j \leftarrow \mathcal{F}_j(\mathrm{C}_j\mathbf{x} + \mathbf{d}_j)$
TV
Pretrained
Neuron-trained
$\mathbf{v}_j$
3 dual update
$\mathbf{d}_j \leftarrow \mathbf{d}_j + \mathrm{C}_j\mathbf{x} - \mathbf{v}_j$